

Vodacom M-Pesa Tanzania

Open API – Complete Developer Documentation

All Endpoints: Authentication • C2B • B2C • B2B • Reversal • Query Status

Version 1.0 | September 2026

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1. Introduction & Base URLs

This document covers the full set of RESTful endpoints exposed by the **M-Pesa Open API** for Vodacom Tanzania. The API allows businesses to collect payments (C2B), disburse funds (B2C/B2B), reverse transactions, and query status.

Base URLs

Sandbox: <https://openapi.m-pesa.com/sandbox/ipg/v2/vodacomTZN/>
Production: <https://openapi.m-pesa.com/openapi/ipg/v2/vodacomTZN/>

All endpoints below are relative to the base URL. Country code is always **TZN** and currency is always **TZS**.

Common Headers (all endpoints)

Header	Value	Required
Authorization	Bearer {Session_Key}	Yes
Content-Type	application/json	Yes
Origin	As configured in portal	Yes

2. Authentication – Get Session Key

Every API call requires a Session Key. Obtain it by encrypting your API Key with the RSA Public Key from the portal, then calling Get Session.

Endpoint

GET /getSession/

Authorization header uses the **encrypted API Key** (not the Session Key). The response contains the Session Key to use for all other calls. Session lifetime is configured in the portal (commonly 24 hours).

Note: Encryption is typically RSA-OAEP with SHA-256. Use the exact Public Key (PEM) downloaded from openapiportal.m-pesa.com.

3. Customer-to-Business (C2B) Single Stage

Request payment from a customer. M-Pesa sends a USSD Push; the customer enters their PIN to authorize. Funds move from customer wallet to your Service Provider Code.

Endpoint

POST /c2bPayment/singleStage/

Request Parameters

Parameter	Type	Req	Description / Example
input_Amount	String	Yes	Amount (whole number). e.g. "1500"
input_CustomerMSISDN	String	Yes	Customer phone (255...). Sandbox: "000000000001"
input_Country	String	Yes	"TZN"
input_Currency	String	Yes	"TZS"
input_ServiceProviderCode	String	Yes	Your short code. Sandbox: "000000"
input_TransactionReference	String	Yes	Your unique ref (≤20 chars). e.g. "ORD-98765"
input_ThirdPartyConversationID	String	Yes	Unique ID from your system (UUID recommended)
input_PurchasedItemsDesc	String	Yes	Description. e.g. "Payment for Order #98765"

Sample Request

```
{
  "input_Amount": "1500",
  "input_CustomerMSISDN": "255712345678",
  "input_Country": "TZN",
  "input_Currency": "TZS",
  "input_ServiceProviderCode": "123456",
  "input_TransactionReference": "ORD-98765",
  "input_ThirdPartyConversationID": "a1b2c3d4-e5f6-7890-abcd-ef1234567890",
  "input_PurchasedItemsDesc": "Payment for Order #98765"
}
```

Sample Success Response

```
{
  "output_ResponseCode": "INS-0",
  "output_ResponseDesc": "Request processed successfully",
  "output_TransactionID": "RvvsqB0rcP3Y",
  "output_ConversationID": "1e029550d09745e7b2221bb4b2dc8ffc",
  "output_ThirdPartyConversationID": "a1b2c3d4-e5f6-7890-abcd-ef1234567890"
}
```

INS-0 only means the USSD Push was accepted. Confirm final success via webhook or Query Status API.

4. Business-to-Customer (B2C)

Disburse funds from your business M-Pesa account to a customer's mobile wallet. Common uses: salaries, refunds, promotions, supplier payments to individuals.

Endpoint

POST /b2cPayment/ (or /b2cPayment/singleStage/ depending on portal version)

Request Parameters

Parameter	Type	Req	Description / Example
input_Amount	String	Yes	Amount to send. e.g. "5000"
input_CustomerMSISDN	String	Yes	Recipient phone (255...). Sandbox: "000000000001"
input_Country	String	Yes	"TZN"
input_Currency	String	Yes	"TZS"
input_ServiceProviderCode	String	Yes	Your short code (debit party). Sandbox: "000000"

input_TransactionReference	String	Yes	Your unique reference
input_ThirdPartyConversationID	String	Yes	Unique conversation ID from your system
input_PaymentItemsDesc	String	Yes	Description. e.g. "Salary Payment - Sept 2026"

Sample Request

```
{
  "input_Amount": "5000",
  "input_CustomerMSISDN": "255712345678",
  "input_Country": "TZN",
  "input_Currency": "TZS",
  "input_ServiceProviderCode": "123456",
  "input_TransactionReference": "SAL-202609-001",
  "input_ThirdPartyConversationID": "b2c-uuid-1234-5678",
  "input_PaymentItemsDesc": "Salary Payment - September 2026"
}
```

Sample Success Response

```
{
  "output_ResponseCode": "INS-0",
  "output_ResponseDesc": "Request processed successfully",
  "output_TransactionID": "4iUThBRRWXMG",
  "output_ConversationID": "f3ca96e7cb7f473a847247d8b887b922",
  "output_ThirdPartyConversationID": "b2c-uuid-1234-5678"
}
```

5. Business-to-Business (B2B)

Transfer funds between two business M-Pesa accounts (short codes). Useful for inter-company settlements, supplier payments to other businesses, and bulk settlements.

Endpoint

POST /b2bPayment/

Request Parameters

Parameter	Type	Req	Description / Example
input_Amount	String	Yes	Amount to transfer. e.g. "10000"
input_Country	String	Yes	"TZN"
input_Currency	String	Yes	"TZS"
input_PrimaryPartyCode	String	Yes	Sender short code (debit). Sandbox: "000000"
input_ReceiverPartyCode	String	Yes	Receiver short code (credit). e.g. "000001"
input_TransactionReference	String	Yes	Your unique reference
input_ThirdPartyConversationID	String	Yes	Unique conversation ID
input_PurchasedItemsDesc	String	Yes	Description. e.g. "Supplier Settlement Q3"

Sample Request

```
{
  "input_Amount": "10000",
  "input_Country": "TZN",
  "input_Currency": "TZS",
  "input_PrimaryPartyCode": "123456",
  "input_ReceiverPartyCode": "654321",
  "input_TransactionReference": "SETTLE-2026-09",
  "input_ThirdPartyConversationID": "b2b-uuid-9876-5432",
  "input_PurchasedItemsDesc": "Supplier Settlement - September"
}
```

Sample Success Response

```
{
  "output_ResponseCode": "INS-0",
  "output_ResponseDesc": "Request processed successfully",
  "output_TransactionID": "8a89835c71f15e99396",
  "output_ConversationID": "c4d5e6f7a8b9c0d1e2f3",
  "output_ThirdPartyConversationID": "b2b-uuid-9876-5432"
}
```

6. Transaction Reversal

Reverse a previously successful transaction. Funds are withdrawn from the recipient and returned to the original initiating party. Partial reversals are supported by specifying a lower amount.

Endpoint

POST /reversal

Request Parameters

Parameter	Type	Req	Description / Example
input_TransactionID	String	Yes	Original M-Pesa TransactionID to reverse
input_ReversalAmount	String	Yes*	Amount to reverse (optional for full reversal in some versions)
input_Country	String	Yes	"TZN"
input_ServiceProviderCode	String	Yes	Your short code
input_ThirdPartyConversationID	String	Yes	Unique ID for this reversal request

* Some portal versions / markets also require **input_SecurityCredential** and **input_InitiatorIdentifier** (provided by Vodacom after B2C/B2B setup). Check your portal application settings.

Sample Request

```
{
  "input_TransactionID": "RvvsqB0rcP3Y",
  "input_ReversalAmount": "1500",
  "input_Country": "TZN",
  "input_ServiceProviderCode": "123456",
  "input_ThirdPartyConversationID": "rev-uuid-1111-2222"
}
```

Sample Success Response

```
{
  "output_ResponseCode": "INS-0",
  "output_ResponseDesc": "Request processed successfully",
  "output_TransactionID": "REV123456XYZ",
  "output_ConversationID": "rev-conv-id-abc",
  "output_ThirdPartyConversationID": "rev-uuid-1111-2222"
}
```

7. Query Transaction Status

Check the current status of any previously initiated transaction using its TransactionID, ConversationID, or ThirdPartyConversationID.

Endpoint

GET or POST /queryTransactionStatus/

Request Parameters

Parameter	Type	Req	Description / Example
input_QueryReference	String	Yes	TransactionID or ConversationID to query
input_ServiceProviderCode	String	Yes	Your short code

input_ThirdPartyConversationID	String	Yes	Unique ID for this query request
input_Country	String	Yes	"TZN"

Sample Request

```
{
  "input_QueryReference": "RvvsqB0rcP3Y",
  "input_ServiceProviderCode": "123456",
  "input_ThirdPartyConversationID": "query-uuid-3333-4444",
  "input_Country": "TZN"
}
```

Sample Success Response

```
{
  "output_ResponseCode": "INS-0",
  "output_ResponseDesc": "Request processed successfully",
  "output_TransactionID": "RvvsqB0rcP3Y",
  "output_ConversationID": "1e029550d09745e7b2221bb4b2dc8ffc",
  "output_ThirdPartyConversationID": "query-uuid-3333-4444",
  "output_ResponseTransactionStatus": "Completed"
}
```

Possible status values: **Completed**, **Pending**, **Cancelled**, **Expired**, **Failed**, **N/A**.

8. Common Response & Error Codes

All endpoints return an **output_ResponseCode** using the INS-xxx series.

Code	Meaning	Action
INS-0	Request processed successfully	Proceed / wait for final status
INS-1	Internal Error	Retry later; contact support if persistent
INS-2	Invalid API Key / Insufficient balance	Check credentials or account balance
INS-4	User not active / Transaction failed	Verify account status
INS-5	Transaction cancelled by customer	Allow customer to retry
INS-6	Transaction Failed / Not permitted	Show failure message
INS-9	Request timeout	Customer did not respond – retry
INS-10	Duplicate transaction	Use new unique references
INS-13	Invalid Shortcode	Verify ServiceProviderCode
INS-14 / INS-17	Invalid Reference	Check length & format (1–20 chars)
INS-15	Invalid Amount	Must be positive whole number
INS-16	Temporary overloading	Retry with backoff
INS-19	Invalid ThirdPartyReference	Ensure uniqueness
INS-20 / INS-21	Missing / invalid parameters	Validate all required fields
INS-26	Not authorized	Check Session Key & permissions
INS-996	Customer account not active	Customer must activate M-Pesa
INS-998	Invalid Market	Use TZN endpoint & Country=TZN

9. Best Practices & Go-Live

- Always generate unique **ThirdPartyConversationID** (UUID) and **TransactionReference** for every request.
- Store ConversationID and TransactionID from synchronous responses for reconciliation and support.
- Configure a secure HTTPS Response URL (webhook) in the portal for asynchronous final results.
- Handle INS-5 (cancelled) and INS-9 (timeout) gracefully with a clear “Retry” option for the user.
- Never hard-code credentials – use environment variables or a secrets manager.
- Fully test in Sandbox (MSISDN **0000000000001**, ServiceProviderCode **000000**) before go-live.
- For production: complete business onboarding with Vodacom M-Pesa Business team and request go-live from the portal.

10. Useful Links & Contacts

- Open API Portal: <https://openapiportal.m-pesa.com>
- Business Onboarding (Tanzania): <https://business.m-pesa.com/vodacom-tanzania/business-onboarding-tanzania/>
- M-Pesa Business email: m-pesabusiness@vodacom.co.tz
- Sandbox test MSISDN: **0000000000001** | Sandbox Short Code: **000000**

This document is a practical developer reference compiled from the official Open API behaviour and widely used community SDKs. Always verify the latest parameter names and response fields inside the Open API Portal after logging in.

Document generated on 2026-09-22 • For official support contact your Vodacom M-Pesa Business Manager